

Fachhochschule Vorarlberg GmbH		Author: Horatiu O. Pilsan	
EIT & MEC - Digi SGL 2. Topic: Sequential Circuits	Rev. 1.1	29.04.2026	Page 1 of 1

1. Objective:

Understand the mode of operation of sequential circuits and get to know the different types.

2. Questions:

1. Which is the mode of operation and the internal structure of simple (clockless) R-S-Flop-Flops?
2. The same question for clock level triggered [taktzustandgesteuerte] Flip-Flops (R-S and D type).
3. The same question for edge triggered [flankengesteuerte] Flip-Flops.

4. Tasks:

1. Sum up the most important pieces of information on the topic on one flip chart without adding too many details! Best suited is a visual manner.
2. Formulate questions on the topic, which you were not able to answer, neither alone nor in the team meeting!

5. Dates:

See course schedule in the information file for the course (_EIT-Digi_Info.pdf).

6. Literature:

- [1] Tietze, Ulrich; Schenk, Christoph (1999): Halbleiter-Schaltungstechnik, 11. Ausgabe, Berlin: Springer, (Kap. 9.1, p. 685 – 692)
- [2] Witowitz, Roland; Urbanski, Klaus; Gehrke, Winfried (2012): Digitaltechnik. Ein Lehr- und Übungsbuch. 6. Auflage. Heidelberg: Springer (Kap. 6.1, p. 175-183).
- [3] Gehrke, Winfried; u.a. (2016): Digitaltechnik. Grundlagen, VHDL, FPGAs, Mikrocontroller, Berlin: Springer-Vieweg (Kap. 5.1, p. 116-126).
- [4] Fricke, Klaus (2014): Digitaltechnik- Lehr- und Übungsbuch für Elektrotechniker und Informatiker, 7. Auflage, Wiesbaden: Springer-Vieweg (Kap. 7.5, p. 78-84).